

## Project Initialization and Planning Phase

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|---------------|-----------------------------------------------------|
| Date          | 26 June 2026                                        |
| Team ID       | SWTID-2026-9014                                     |
| Project Title | AdForge AI: AI-Powered Advertisement Copy Generator |
| Maximum Marks | 5 Marks                                             |

### Project Proposal (Proposed Solution) report

The proposal report aims to transform digital advertisement creation using generative AI, boosting marketing efficiency and copy quality. It tackles the inefficiencies of manual ad writing and the high cost of professional copywriters, promising faster campaign launches, reduced marketing costs, and higher conversion rates. Key features include a multi-platform AI advertisement generator powered by Qwen3-32B, a real-time ad refinement engine, and an intelligent performance tips system.

| Project Overview  |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| Objective         | The primary objective of AdForge AI is to revolutionize the digital advertisement creation process by implementing advanced generative AI techniques, enabling businesses, marketers, startups, and agencies to produce high-converting, platform-specific advertisement copy within seconds — eliminating the need for expensive copywriters or generic AI tools.                                                              |
| Scope             | The project comprehensively addresses the end-to-end advertisement creation workflow, incorporating a large language model (Qwen3-32B via Groq API) to generate platform-native ad copy for Google Ads, Instagram, and Facebook. The scope includes multi-platform ad generation, marketing slogan creation, advertisement refinement, and performance optimization tips — all accessible through a single web-based interface. |
| Problem Statement |                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| Description              | Small businesses, startups, and marketing professionals struggle to create high-quality, platform-specific advertisement copy due to lack of copywriting expertise, high cost of professional copywriters, and the inability of generic AI tools to understand platform-specific requirements such as Google Ads character limits, Instagram hashtag culture, and Facebook's conversational tone.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Impact                   | Solving these issues will result in faster campaign launches, reduced marketing costs, improved advertisement performance across platforms, and increased accessibility of professional-grade marketing for businesses of all sizes — directly contributing to higher click-through rates, better conversions, and greater return on ad spend.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Proposed Solution</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Approach                 | <p>Employing a state-of-the-art large language model (Qwen3-32B) accessed via the Groq API to analyze campaign briefs and generate psychologically optimized, platform-native advertisement copy. The system applies proven marketing frameworks including AIDA, PAS, FAB, BAB, FOMO, Scarcity, Social Proof, and Urgency — automatically selecting the most appropriate techniques based on the product, audience, and campaign goal provided by the user.</p> <p><b>Key Features</b></p> <ul style="list-style-type: none"> <li>• Multi-platform ad generation for Google Ads, Instagram, and Facebook with platform-specific formatting and character limits</li> <li>• Marketing slogan generator producing 5 unique, brandable slogans per campaign</li> <li>• Advertisement Refinement Engine that improves existing ads based on user feedback with improvement scoring</li> <li>• Performance Tips Generator providing A/B testing ideas, CTA improvements, SEO suggestions, and audience recommendations</li> <li>• Support for 16 marketing tones including Professional, Luxury, Urgent, Emotional, Playful, and Youthful</li> <li>• Input safety filter rejecting harmful, fraudulent, or inappropriate content</li> <li>• Copy-to-clipboard and download as TXT for instant use</li> </ul> |
| Key Features             | <ul style="list-style-type: none"> <li>• Implementation of a generative AI-based advertisement copy engine powered by Qwen3-32B via Groq API.</li> <li>• Real-time multi-platform ad generation for Google Ads, Instagram, and Facebook with platform-specific formatting and character limits.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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|  | <ul style="list-style-type: none"> <li>• Advertisement Refinement Engine that improves existing ad copy based on user feedback with improvement scoring and conversion prediction.</li> <li>• Marketing slogan generator producing 5 unique, memorable, and brandable slogans per campaign.</li> <li>• Performance Tips Generator providing A/B testing ideas, CTA improvements, SEO suggestions, best posting times, and audience recommendations.</li> <li>• Support for 16 marketing tones including Professional, Luxury, Urgent, Emotional, and Playful to match any brand voice.</li> <li>• Built-in content safety filter that automatically rejects harmful, fraudulent, or inappropriate advertisement requests.</li> <li>• Copy-to-clipboard and download as TXT functionality for instant use of generated advertisements.</li> </ul> |
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## Resource Requirements

| Resource Type                                                       | Description                             | Specification/Allocation                                        |
|---------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• <b>Hardware</b></li> </ul> |                                         |                                                                 |
| Computing Resources                                                 | CPU/GPU specifications, number of cores | Intel Core i5 8th Gen or above / AMD Ryzen 5                    |
| Memory                                                              | RAM specifications                      | Minimum 8 GB RAM                                                |
| Internet                                                            | Speed of the internet                   | Minimum 10 Mbps stable connection (required for Groq API calls) |
| Storage                                                             | Disk space for data, models, and logs   | 256 GB SSD minimum                                              |
| <b>Software</b>                                                     |                                         |                                                                 |

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|-------------------------|------------------------------------------------|-----------------------------------------------------------------|
| Operating System        | The main software                              | Windows 10/11, macOS Monterey+, Ubuntu 20.04+                   |
| Libraries               | Additional libraries                           | groq==0.9.0, python-dotenv==1.0.1, bleach==6.1.0, httpx==0.27.0 |
| Backend Framework       | Python Framework                               | Python Flask 3.0.3                                              |
| AI Model                | To recognize patterns and generate new content | Qwen3-32B (accessed via Groq API)                               |
| Frontend                | Visual part of the application                 | HTML5, CSS3, Vanilla JavaScript                                 |
| Development Environment | IDE                                            | Visual Studio Code, PowerShell terminal                         |